Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 5 Solutions

Problem 5.1

A transmitter sends data packets over a computer network. Let T be the travel time of a packet and S be its size. A packet contains 10^2 bytes with probability 5/6 and 10^4 bytes with probability 1/6. The travel time T of the data packet depends on its size S and on the congestion in the network at the time of transmission. In particular, the travel time is $10^{-3} \cdot S$ seconds with probability 1/3 and $10^{-1} \cdot S$ seconds with probability 2/3.

(a) Determine the PMF of S.

Solution: From the problem description it follows that

$$p_S(s) = \begin{cases} 5/6 & s = 10^2 \\ 1/6 & s = 10^4. \end{cases}$$

(b) Determine the conditional PMF of T given S for each value of S.

Solution: From the problem definition it follows that

$$p_{T|S}(t|10^2) = \begin{cases} 1/3 & t = 10^{-1} \\ 2/3 & t = 10. \end{cases}$$

and

$$p_{T|S}(t|10^4) = \begin{cases} 1/3 & t = 10 \\ 2/3 & t = 10^3. \end{cases}$$

(c) Determine the marginal PMF of T.

Solution: We use the Total Probability Law to obtain

$$\begin{aligned} p_T(t) &= \sum_{s \in \{10^2, 10^4\}} p_S(s) p_{T|S}(t|s) \\ &= \begin{cases} \frac{5}{6} \cdot \frac{1}{3} = \frac{5}{18} & t = 10^{-1} \\ \frac{5}{6} \cdot \frac{2}{3} + \frac{1}{6} \cdot \frac{1}{3} = \frac{11}{18} & t = 10 \\ \frac{1}{6} \cdot \frac{2}{3} = \frac{1}{9} & t = 10^3. \end{cases} \end{aligned}$$

We can easily verify that $\sum_{t \in \{10^{-1}, 10, 10^3\}} p_T(t) = 1$.

(d) Determine the expected travel time for a data packet chosen at random.

Solution: One approach is to use the PMF of T we computed in part (c) to obtain

$$\begin{aligned}\mathbb{E}[T] &= \sum_{t\in\{10^{-1},10,10^3\}} p_T(t)\, t \\ &= \frac{5}{18}\cdot 10^{-1} + \frac{11}{18}\cdot 10 + \frac{1}{9}\cdot 10^3 = 117.25\,\text{sec}.\end{aligned}$$

Alternatively, we can use the iterated expectations to compute $\mathbb{E}[T]$. Using the conditional PMFs we derived in part (b), we obtain

$$\begin{aligned}\mathbb{E}\big[T|S=10^2\big] &= \frac{1}{3}\cdot 10^{-1} + \frac{2}{3}\cdot 10 = 6.7\,\text{sec} \\ \mathbb{E}\big[T|S=10^4\big] &= \frac{1}{3}\cdot 10 + \frac{2}{3}\cdot 10^3 = 670\,\text{sec} \\ \mathbb{E}[T] &= \frac{5}{6}\cdot\mathbb{E}\big[T|S=10^2\big] + \frac{1}{6}\cdot\mathbb{E}\big[T|S=10^4\big] = 117.25\,\text{sec}.\end{aligned}$$

(e) We define $U \triangleq S/T$ to be the bandwidth at which the packet was transmitted. Determine the expected bandwidth of the data packets that are transmitted over this network.

Solution: We define $g(s,t) \triangleq s/t$ and use the formula for the expectation of a function of random variables to compute

$$\begin{aligned}\mathbb{E}[U] &= \sum_{s,t} p_{S,T}(s,t)\, g(s,t) \\ &= \sum_{s,t} p_S(s)\, p_{T|S}(t|s)\, g(s,t) \\ &= \frac{5}{6}\cdot\left(\frac{1}{3}\cdot 10^3 + \frac{2}{3}\cdot 10\right) + \frac{1}{6}\cdot\left(\frac{1}{3}\cdot 10^3 + \frac{2}{3}\cdot 10\right) \\ &= \frac{1}{3}\cdot 10^3 + \frac{2}{3}\cdot 10 = 340\,\text{bits/sec},\end{aligned}$$

where to obtain the second equality we have used the conditional distribution definition and to obtain the third equality we have used the conditional distributions we derived in part (b).

Remark: Note that in the computation of the expectation above, the conditional expectation of U given S was the same for both values of S. Moreover, the conditional distribution of U given S does not depend on the specific value of S. We will soon see that this corresponds to the notion of *independence* of random variances U and S, which is closely related to the independence of events we have seen in the course.

Problem 5.2

Let $X \in \mathcal{X} = \{\texttt{rainy}, \texttt{sunny}\}$ be the weather in Massachusetts. Assume X is equally likely to be rainy (`rainy`) or sunny (`sunny`).

(a) Determine the PMF and entropy of X.

Solution: From the problem description it follows that

$$p_X(x) = \frac{1}{2}, \quad x = \texttt{rainy}, \texttt{sunny}.$$

Since X is a uniformly distributed random variable,

$$H(X) = \log |\mathcal{X}| = \log 2 = 1 \text{ bit}.$$

Let $Y \in \mathcal{X}$ be the weather in New Hampshire. Since the two states are close, if the weather in Massachusetts is $X = x$, then the weather in New Hampshire is $Y = x$ with probability $1 - \epsilon$, where $\epsilon < 1/2$ is a known small positive constant.

(b) Determine the joint PMF of X and Y.

Solution: The problem setup implies

$$p_{Y|X}(y|x) = \begin{cases} 1 - \epsilon & y = x \\ \epsilon & y \neq x. \end{cases}$$

We use the definition of the conditional probability to obtain

$$\begin{aligned} p_{X,Y}(x, y) &= p_X(x)\, p_{Y|X}(y|x) \\ &= \begin{cases} \dfrac{1-\epsilon}{2} & y = x \\ \dfrac{\epsilon}{2} & y \neq x. \end{cases} \end{aligned}$$

(c) Determine the distribution of the weather in Massachusetts given that it is a sunny day in New Hampshire.

Solution: Using the Bayes' rule, we compute

$$\begin{aligned} p_{X|Y}(x|\texttt{sunny}) &= \frac{p_X(x)\, p_{Y|X}(\texttt{sunny}|x)}{p_X(\texttt{rainy})\, p_{Y|X}(\texttt{sunny}|\texttt{rainy}) + p_X(\texttt{sunny})\, p_{Y|X}(\texttt{sunny}|\texttt{sunny})} \\ &= \frac{\frac{1}{2} \cdot p_{Y|X}(\texttt{sunny}|x)}{\frac{1}{2} \cdot \epsilon + \frac{1}{2} \cdot (1 - \epsilon)} \\ &= p_{Y|X}(\texttt{sunny}|x) \\ &= \begin{cases} \epsilon & x = \texttt{rainy} \\ 1 - \epsilon & x = \texttt{sunny}, \end{cases} \end{aligned}$$

i.e., knowing that it is sunny in New Hampshire changes our belief about the weather in Massachusetts. Specifically, the probability that it is sunny increases from $p_X(\texttt{sunny}) = 1/2$ to $p_{X|Y}(\texttt{sunny}|\texttt{sunny}) = 1 - \epsilon > 1/2 = p_X(\texttt{sunny})$.

Remark: Due to the symmetry in this problem, $p_{X|Y}(x|y) = p_{Y|X}(y|x)$ for all $x, y \in \mathcal{X}$, which is not true in general.

(d) Determine the conditional entropy of X given Y.

Solution: We first observe that given $Y = y$, the distribution of X is a Bernoulli distribution with parameter ϵ, and therefore

$$H(X|Y = \texttt{rainy}) = H(X|Y = \texttt{sunny}) = H_{\texttt{B}}(\epsilon).$$

We now use iterated expectation to compute

$$\begin{aligned} H(X|Y) &= \sum_{y \in \mathcal{X}} p_Y(y)\, H(X|Y = y) \\ &= \sum_{y \in \mathcal{X}} p_Y(y)\, H_{\texttt{B}}(\epsilon) \\ &= H_{\texttt{B}}(\epsilon) \\ &= -\epsilon \log \epsilon - (1 - \epsilon) \log(1 - \epsilon). \end{aligned}$$

As we saw in class, $H_{\texttt{B}}(\epsilon) < H_{\texttt{B}}(1/2)$, thus the entropy of X decreases on average after observing Y. In this case, the entropy decreases for every value of Y, which is not always the case.